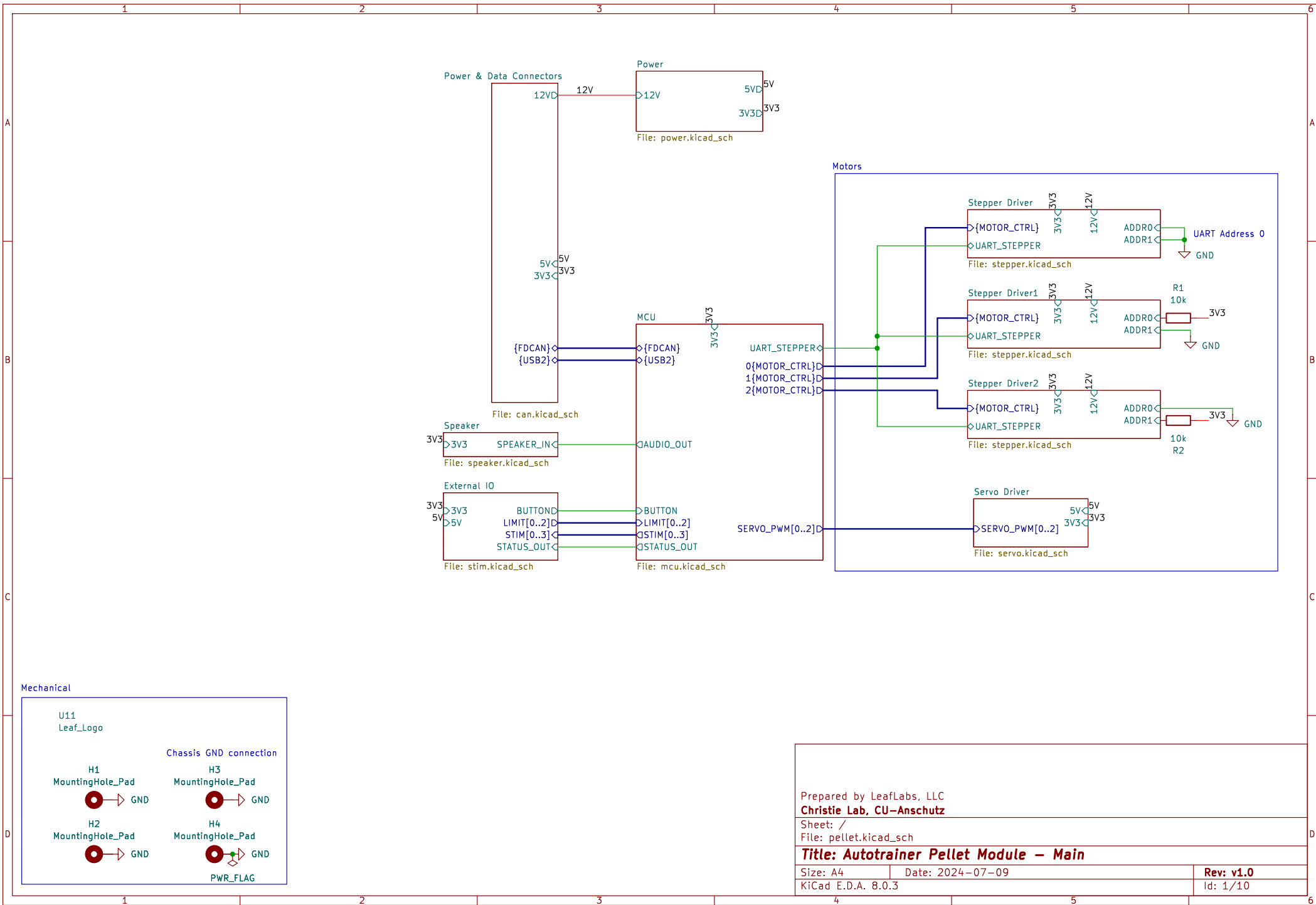
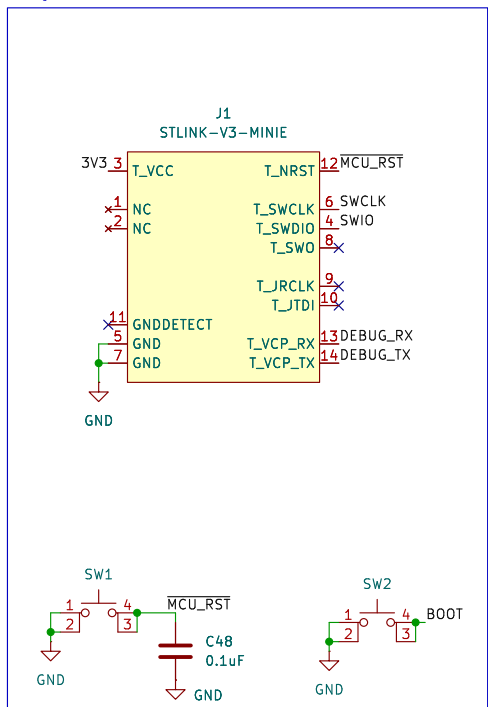
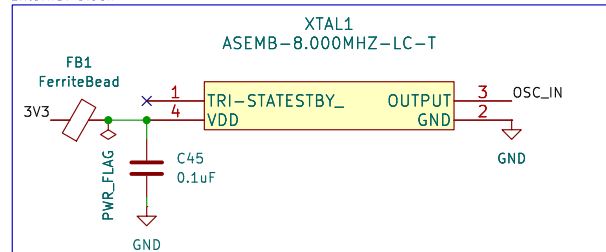




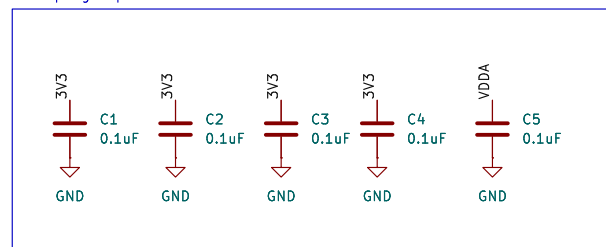
# Debug



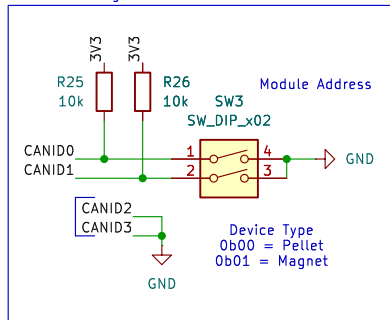
# External clock



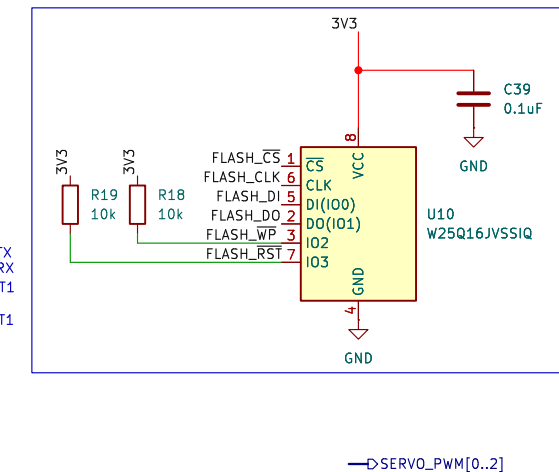
# Decoupling Capacitor Central



# CAN Addressing



# SPI Flash



Prepared by LeafLabs, LLC  
Christie Lab, CU-Anschutz

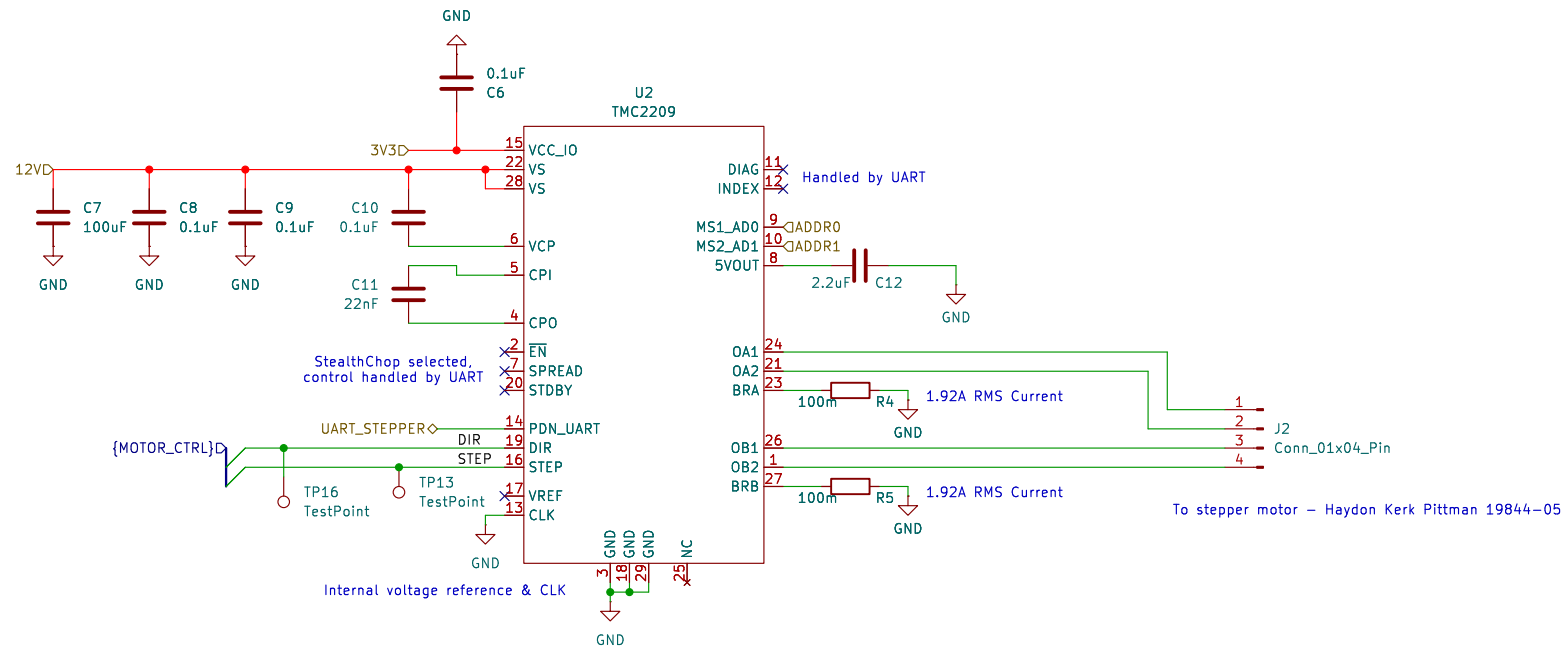
Sheet: /MCU/  
File: mcu.kicad\_sch

Title: Autotrainer Pellet Module – MCU

Size: A4  
KiCad E.D.A. 8.0.3

Date: 2024-07-09

Rev: v1.0  
Id: 2/10



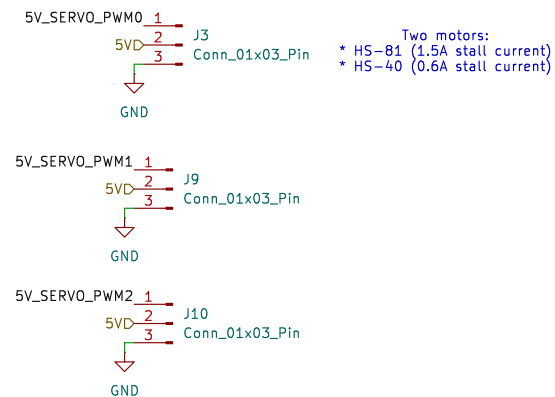
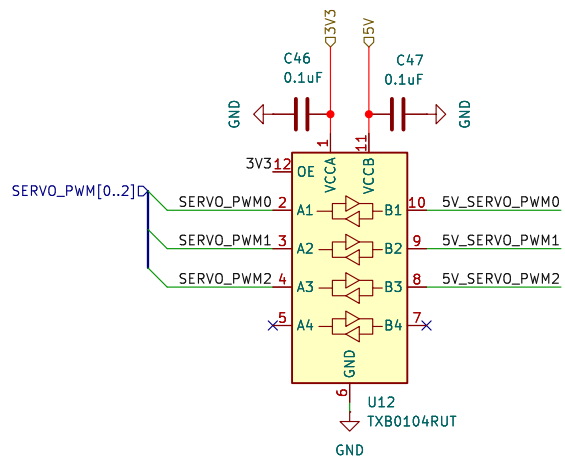
Prepared by LeafLabs, LLC  
**Christie Lab, CU–Anschutz**

Sheet: /Stepper Driver/  
 File: stepper.kicad\_sch

**Title: Autotrainer Pellet Module – Stepper**

|                    |                  |           |
|--------------------|------------------|-----------|
| Size: A4           | Date: 2024–07–09 | Rev: v1.0 |
| KiCad E.D.A. 8.0.3 |                  | Id: 3/10  |

# Servo Level Shifting



Prepared by LeafLabs, LLC  
**Christie Lab, CU-Anschutz**

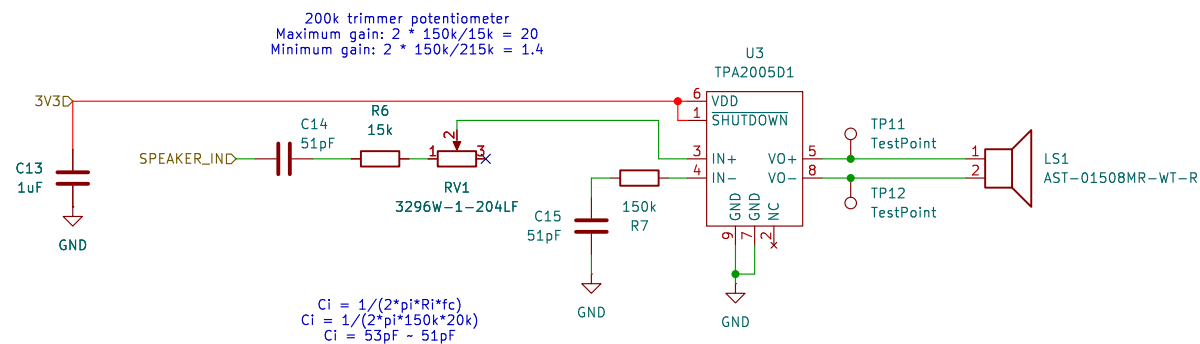
Sheet: /Servo Driver/  
File: servo.kicad\_sch

**Title: Autotrainer Pellet Module – Servo**

Size: A4  
KiCad E.D.A. 8.0.3

Date: 2024-07-09

Rev: v1.0  
Id: 4/10



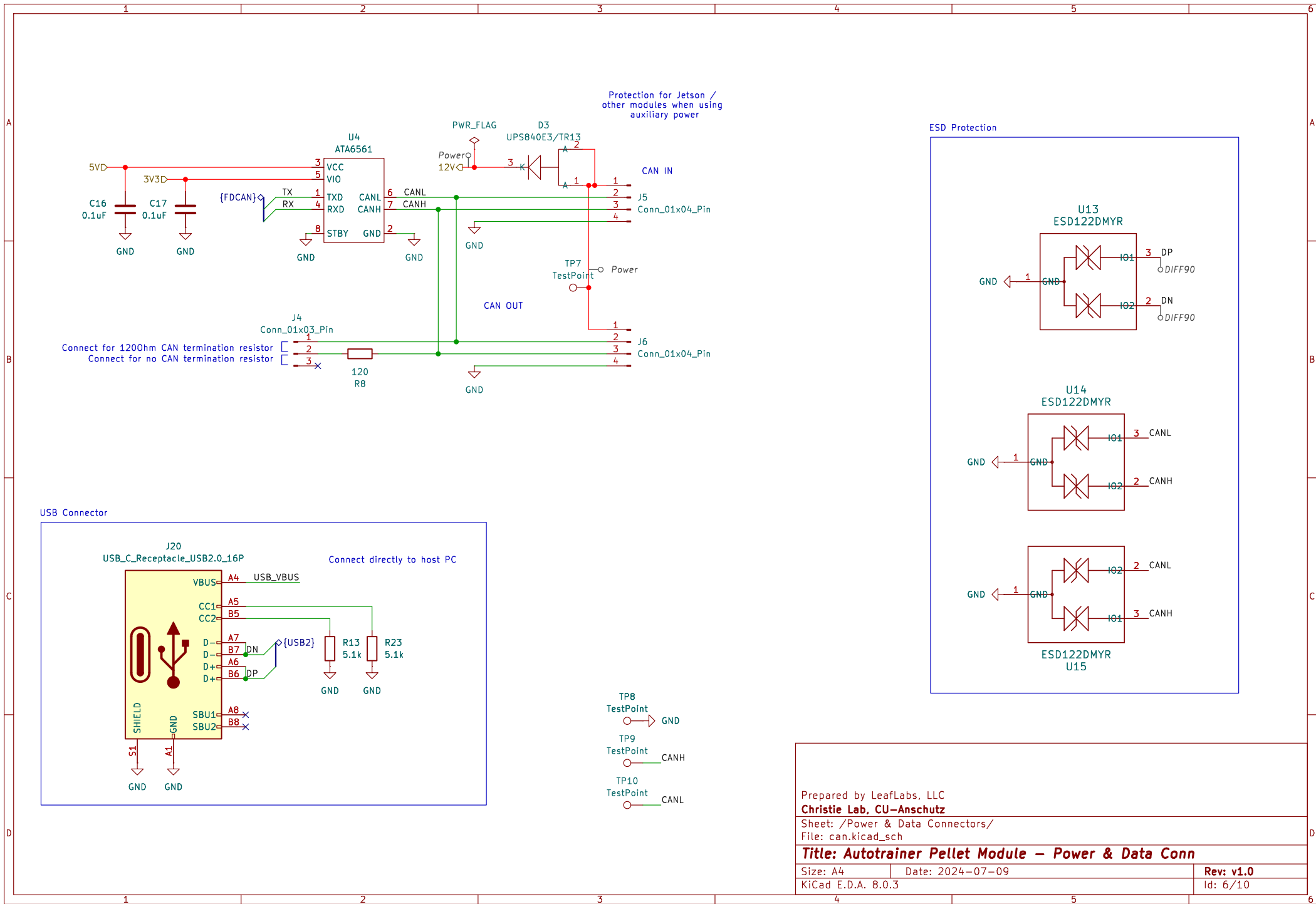
Mouse audible range is 1k-91kHz  
Human audible range is 20Hz-20kHz  
Design for audio between 1k-20kHz

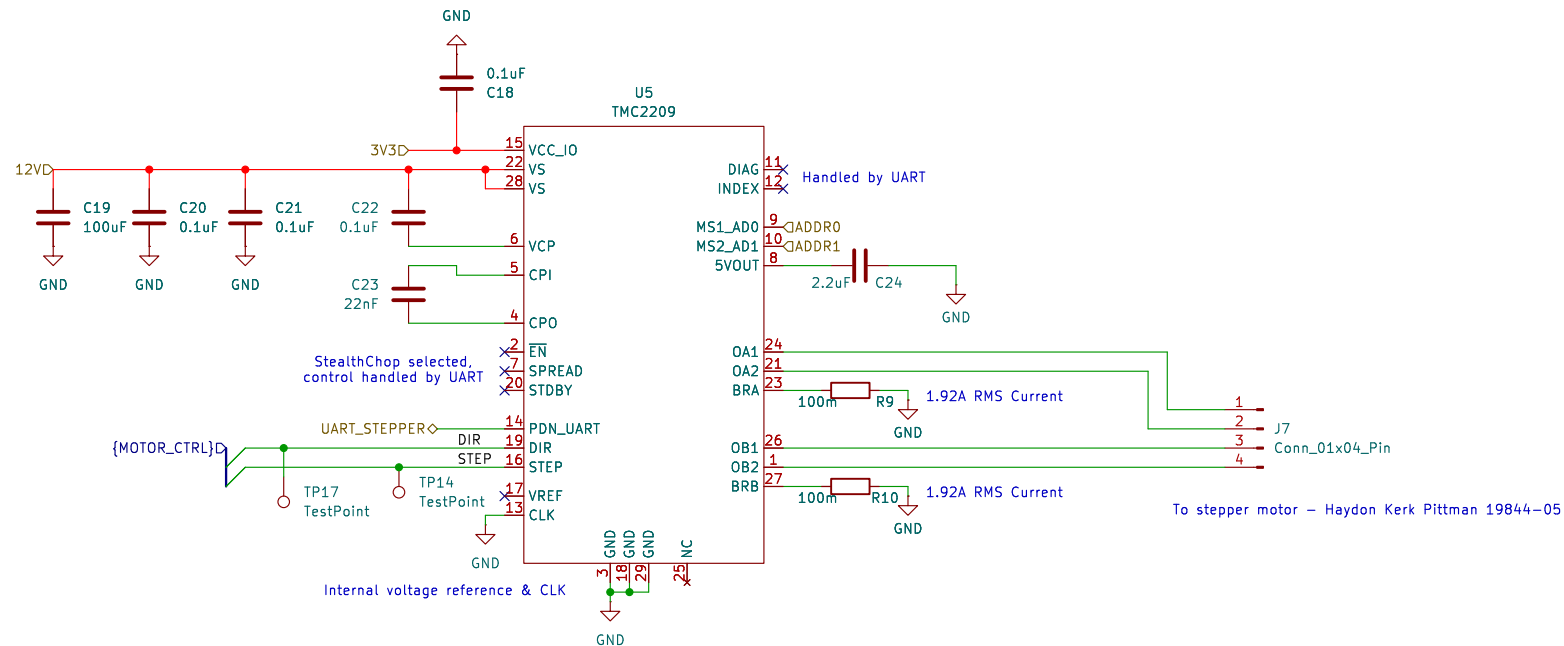
Prepared by LeafLabs, LLC  
**Christie Lab, CU-Anschutz**

Sheet: /Speaker/  
File: speaker.kicad\_sch

**Title: Autotrainer Pellet Module – Speaker**

|                    |                  |           |
|--------------------|------------------|-----------|
| Size: A4           | Date: 2024-07-09 | Rev: v1.0 |
| KiCad E.D.A. 8.0.3 |                  | Id: 5/10  |





Prepared by LeafLabs, LLC  
**Christie Lab, CU-Anschutz**

Sheet: /Stepper Driver1/  
 File: stepper.kicad\_sch

**Title: Autotrainer Pellet Module – Stepper**

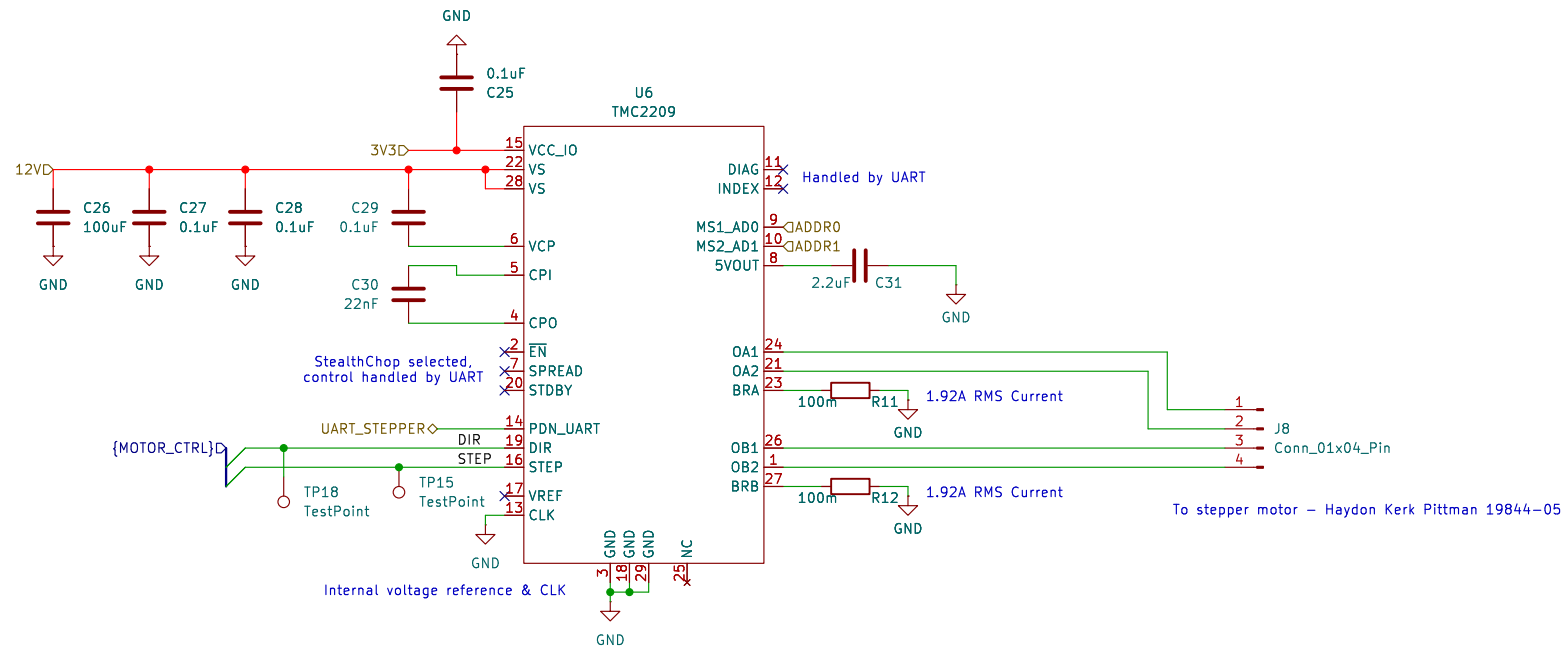
Size: A4

Date: 2024-07-09

Rev: v1.0

KiCad E.D.A. 8.0.3

Id: 7/10



Prepared by LeafLabs, LLC  
**Christie Lab, CU–Anschutz**

Sheet: /Stepper Driver2/  
 File: stepper.kicad\_sch

**Title: Autotrainer Pellet Module – Stepper**

Size: A4

Date: 2024–07–09

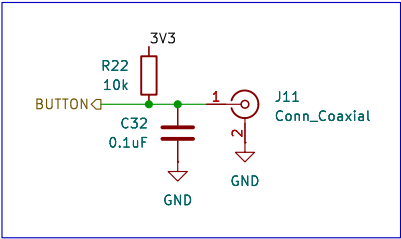
Rev: v1.0

KiCad E.D.A. 8.0.3

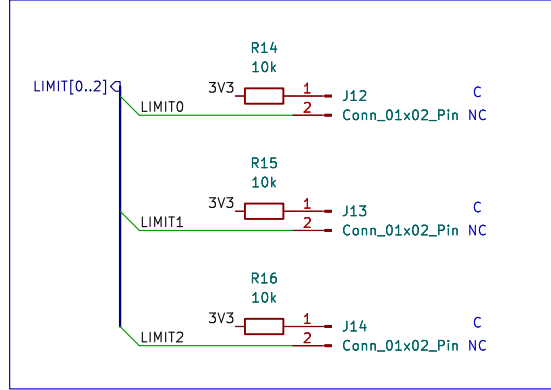
Id: 8/10



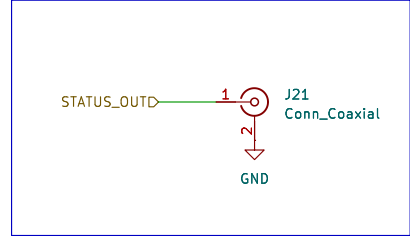
Connection to external button



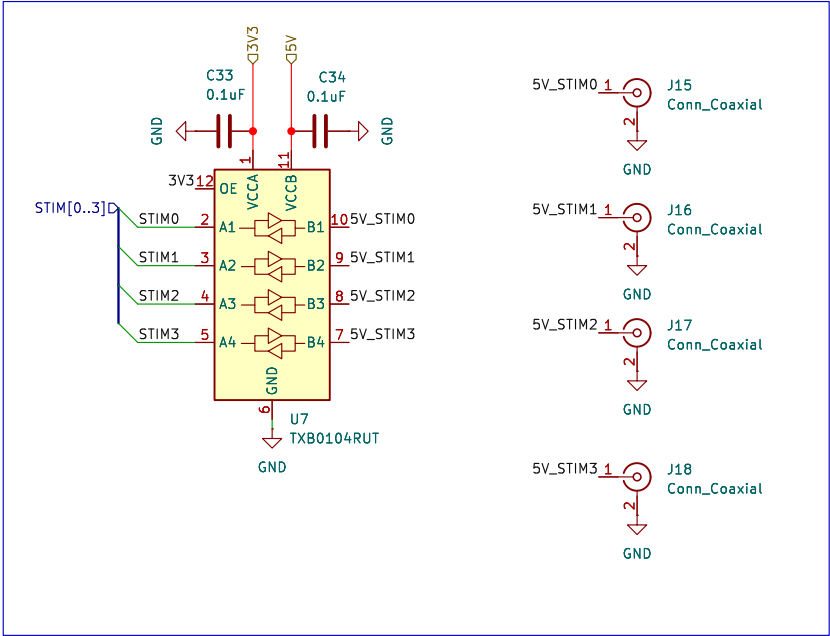
Connections to Limit Switches



Analog status indicator



External Equipment



Prepared by LeafLabs, LLC  
**Christie Lab, CU-Anschutz**

Sheet: /External IO/  
File: stim.kicad\_sch

**Title: Autotrainer Pellet Module – External IO**

Size: A4

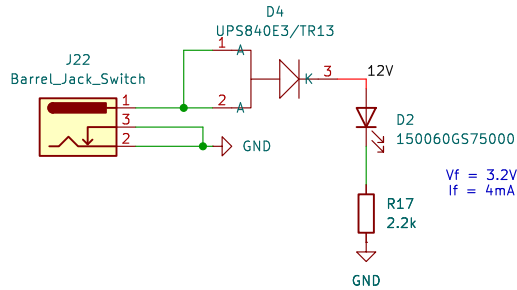
Date: 2024-07-09

Rev: v1.0

KiCad E.D.A. 8.0.3

Id: 11/10

# Aux Power In



12V to 5V converter  
Max expected current consumption: 4.2 A

Inductor selection math

$$I_{pp} = V_{out}/V_{in(max)} * (V_{in(max)} - V_{out})/(L_o * f_{sw})$$

$$I_{pp} = 5/12.1 * (12.1 - 5)/(3.3\mu H * 550 \text{ kHz})$$

$$I_{pp} = 1.6 \text{ A}$$

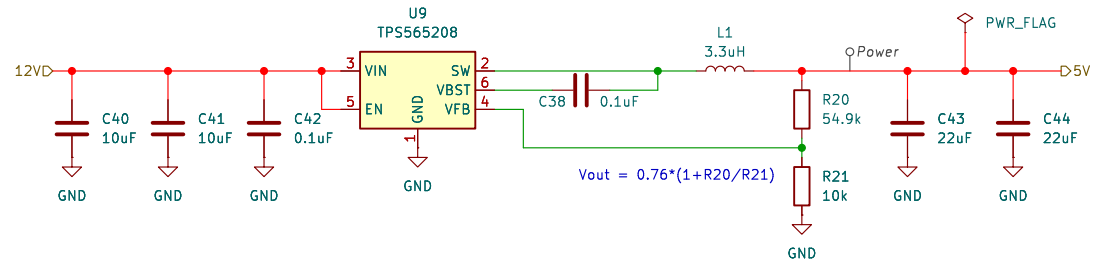
Capacitor current

$$I_{co(rms)} = (V_{out} * (V_{in} - V_{out})) / (\sqrt{12} * V_{in} * L * f_{sw})$$

$$I_{co(rms)} = 1.11 \text{ A}$$

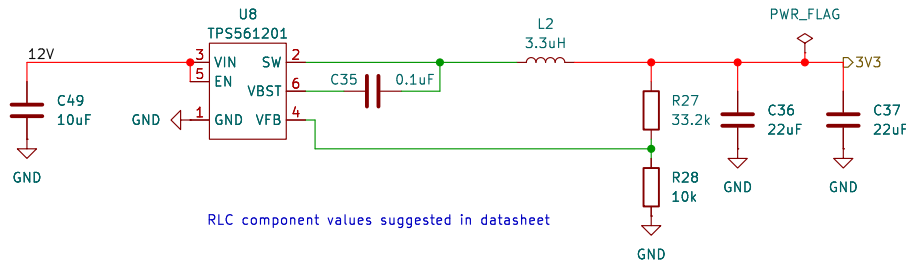
$$I_{peak} = I_o + I_{pp}/2 = 4.2 + 1.6 = 5.8 \text{ A}$$

$$I_o(rms) = \sqrt{I_o^2 + 1/12 I_{pp}^2} = 4.23 \text{ A}$$



12V to 3V3 Converter

Expected current consumption: 126mA  
Supports max 500mA



RLC component values suggested in datasheet

$$I_{pp} = 3.3/12.1 * (12.1 - 3.3)/(3.3\mu H * 580\text{kHz}) = 1.25 \text{ A}$$

$$I_{peak} = 0.5 + 1.25/2 = 1.125 \text{ A}$$

$$I_{to(rms)} = \sqrt{0.5^2 + 1/12 I_{pp}^2} = 0.62 \text{ A}$$

TP1  
TestPoint  
12V

TP2  
TestPoint  
5V

TP3  
TestPoint  
3V3

TP4  
TestPoint  
GND

TP5  
TestPoint  
GND

TP6  
TestPoint  
GND

Prepared by LeafLabs, LLC  
Christie Lab, CU-Anschutz

Sheet: /Power/  
File: power.kicad\_sch

Title: Autotrainer Pellet Module – Power

Size: A4  
Date: 2024-07-09  
KiCad E.D.A. 8.0.3

Rev: v1.0  
Id: 12/10